Tensor computations

Rayleigh–Ritz discriminant degrees for rank-one matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: Open

Rating rationale: Challenging because general discriminant-degree formulas must account for degeneracies and the excluded isotropic locus; specialist importance concerns algebraic conditioning of rank-one Rayleigh–Ritz optimization.
Source: Borovik–Friedman–Hoşten–Pfeffer, Conjecture 3.18 in v2.

Problem statement

For integers $m, n \geq 2$, put $N = mn$ and let

$$V_{m,n} = \{[\mathrm{vec}(ab^T)] : 0 \neq a \in \mathbb{C}^m,\ 0 \neq b \in \mathbb{C}^n\} \subset \mathbb{P}^{N-1}.$$

This is the Segre variety of nonzero rank-one $m \times n$ matrices modulo scaling, in the ordinary entry coordinates. For a complex symmetric matrix $H \in \mathbb{C}^{N\times N}$ define

$$R_H([\psi]) = \frac{\psi^T H \psi}{\psi^T \psi}, \qquad [\psi] \in V_{m,n}, \quad \psi^T\psi \neq 0.$$

Transpose here is bilinear transpose, not conjugate transpose. A critical point is one at which the differential of R_H restricted to the smooth variety $V_{m,n}$ vanishes. It is degenerate if its Hessian in local coordinates on $V_{m,n}$ is singular; at a critical point this condition is independent of the local coordinates.

Let $\Delta^{\mathrm{ni}}_{m,n}$ be the Zariski closure in $\mathbb{P}(\mathrm{Sym}^2\,\mathbb{C}^N)$ of the set of nonzero symmetric matrices H, modulo scaling, for which such a degenerate critical point exists with $\psi^T\psi \neq 0$. This is the source's nonisotropic discriminant.

Conjecture. For every integer $n \geq 2$, these discriminants are hypersurfaces and their degrees are

$$\deg \Delta^{\mathrm{ni}}_{2,n} = 24\binom{n+1}{3}, \qquad \deg \Delta^{\mathrm{ni}}_{3,n} = 24n^2\binom{n}{2}.$$

Degree means projective degree of the reduced hypersurface, equivalently its intersection number with a general projective line. Both formulas are retained as one source conjecture.

Why it matters in numerical linear algebra

Minimizing R_H over real rank-one matrix states is a constrained energy problem underlying tensor variational algorithms. The discriminant records degeneracies of stationary states, where local solution branches can meet and conditioning can deteriorate.

References

1. Viktoriia Borovik, Hannah Friedman, Serkan Hoşten, and Max Pfeffer, *Numerical Algebraic Geometry for Energy Computations on Tensor Train Varieties*, arXiv:2512.06939v2 (2026), §3.2, Proposition 3.7, definition before Example 3.11, Conjecture 3.18; §7.2, Table 4.
2. Flavio Salizzoni, Luca Sodomaco, and Julian Weigert, *Nonlinear Rayleigh quotient optimization*, 2025, §§2 and 5.

Status check — 2026-09-10

Rechecked Borovik et al. v2, Conjecture 3.18 and Table 4, and searched the title, authors, discriminant and conjecture number. The June 2026 revision retains both formulas as one conjecture. Small-dimensional numerical evidence and generic critical-point counts are not proofs of the stated general discriminant degrees. No later resolution was located.